1: Select any system (eg. ATM System) and thoroughly examine its System Specification to identify and document the different software defects prevent.

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Features to be tested:

1. Vality of the ATM card
2. Withdrawal transcation flow of the ATM
3. Authentication of the User
4. Dispersing cash form the account
5. Account balance verification
6. Withdrawal limit per transaction
7. Daily withdrawal limit
8. Valid money denominations
9. Receipt printing

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Bug ID	Bug Name
ATM 001	Invalid Card
ATM 002	Invalid PIN
ATM 003	Invalid Account type.
ATM 004	Insufficient Balance
ATM 005	Transcation limit Exceeded
ATM 006	Daily Withdrawal Limit Exceeded
ATM 007	Invalid Money Denominations
ATM 008	Receipt Not Printed
ATM 009	Pin Change Mismatch.

Bug report:

Bug ID: ATM 001

Bug Description: Invalid Card

Steps to reproduce: 3 Insert an invalid or Unrecognized card into ATM.

Expected Result: ATM Should Display "Invalid Card" message.

Actual Result: ATM Displays Welcome Screen if defects Occur

Status: Pass/Fail.

Bug ID: ATM 002

Bug Desc: Invalid Pin

Steps to Reproduce: 3. Insert valid ATM Card

4. Enter an Incorrect Pin

5. Submit the PIN

Expected Result: ATM Should Display "Invalid PIN" & deny access

Actual Result: ATM Displays menu screen, if defects Occur

Status: Pass/Fail.

Bug ID: ATM 003

Bug Desc: Invalid Account Type

Steps to Reproduce: 3. Insert a valid ATM Card and enter Correct PIN

4. Select the withdrawal option from main menu.

5. Select an Unsupported or invalid account type.

Expected Result: ATM Should reject invalid account types, display appropriate error message.

Actual Result: ATM displays amount- entry Screen, if defect occurs

Status: Pass/Fail.

Bug ID: ATM 004

Bug Desc: Insufficient Balance

Steps to Reproduce:

3. Insert a valid ATM Card, enter correct PIN
4. Select Withdrawal Option
5. Select valid account type.
6. enter amount greater than available balance.
7. Confirm the transaction

Expected Result: ATM should "Insufficient Balance" message, reject the transaction.

Actual Result: ATM displays cash collection screen, deducts the amount, if defect occurs.

Status: Pass/Fail

Bug ID: ATM 005

Bug Desc: Withdrawal Limit per Transaction

Steps to Reproduce:

3. Insert a valid ATM card, enter correct PIN.
4. Select Withdrawal Option.
5. Select a valid account type
6. Enter an amount within permitted transaction limit
7. Confirm transaction.

Expected Result: ATM should dispense cash, provide receipt, if Sufficient Balance.

Actual Result: ATM displays a "Transcation limit Exceeded" message if defect occurs.

Status: Pass/Fail.

2. Arms portal - positive test scenario & test cases

Positive Test Scenario:

Test Case ID: TC01

* Test Case: Login with valid credentials
* Precondn: User must have registered account
* Steps to execute:
    4. Open arms portal
    5. Enter valid Username
    6. Enter valid password
    7. Click Login Button
* Test Data: Valid Username and password
* Expected Result: User should log in Successfully, dashboard displayed.
* Status: Pass/Fail.

Test Case ID: TC02

* Test Case: View Student profile
* Precondn: Student must be logged on
* Steps to Execute:
    4. Log in to portal
    5. Open profile section
    6. View student details
* Test Data: Registered student account
* Expected Result: Correct student name, register name, department, other profile details
* Status: Pass/Fail

Test Case ID: TC03

* Test Case: View attendance details
* Precondn: Students must be logged in & attedance record must available.
* Steps to Execute:
    4. Log in to the portal
    5. Open Attendance section
    6. Select year sem or subj
* Test Data: Valid Student account with attendance records
* Expected Result: Correct subj-wise attendance, attendance %, shld displayed
* Status: Pass/Fail

Test Case ID: TC04

* Test Case: View Internal marks
* Precondn: Internal marks must be published
* Steps to Execute:
    4. Log in to portal
    5. Open marks section
    6. Select year subj or semester.
* Test Data: Student account with published marks
* Expected Result: Correct Internal marks displayed
* Status: Pass/Fail.

Test Case ID: TC05

* Test Case: View class timetable
* Precondition: Timetable must be available
* Steps to Execute:
    4. Log into Portal
    5. Open Timetable section
    6. Select current semester

* Test Data: Valid Student account

* Expected Result: Correct class Schedule, subj, timings should display

* Status: Pass/Fail.

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3) Arms Portal - Negative Test Scenarios & Test Cases

Negative Test Cases:

Test Case ID: NTC1

* Test Case: Login with incorrect password

* Precondn: User must have registered account

* Steps to Execute:

4. Open ARMS portal
5. Enter valid Username
6. Enter incorrect password
7. Click Login

* Test Data: Valid Username & Incorrect Password

* Expected Result: Login should fail, appropriate error message

* Status: Pass/Fail.

Test Case ID: NTC2

* Test Case: Login with Unregistered username

* Precondn: ARMS portal login page must available

* Steps to Execute:

4. Open the portal
5. Enter an Unregistered username
6. Enter a password
7. Click Login

* Test Data: Unregistered Username

* Expected Result: System should reject login attempt

* Status: Pass/Fail

Test Case ID: NT03

* Test Case: Submit login form with empty fields

* Precondn: Login page must displayed

* Steps to Execute:

    4. Open the portal

    5. Leave Username, password fields Empty

    6. Click login

* Test Data: Empty Username, password

* Expected Result: System should display req-field validation & login

* Status: Pass/Fail.

Test case ID: NT04

* Test Case: Enter invalid register number

* Precondition: Submit registration (or) search form

* Steps to Execute:

    1. Open Student details form

    2. Enter invalid register no

    3. Click Submit or search

* Test Data: @@@#####

* Expected Result: System Should reject invalid register no, display messg

* Status: Pass/Fail

Test Case ID: NT05

* Test Case: Access another students academic records
* Precondition: Student must be logged in
* Steps to Execute:
    1. Login using Student account
    2. Attempt access another students record by modifying Student ID in request or URL
    3. Submit the request.
* Test Data: Another Students ID
* Expected Results: System should deny access, protect students info
* Status: Pass/Fail

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4) E-commerce Application - Positive test Scenario & test Cases

Positive Test Cases:

Test Case ID: ETC01

* Test Case: Register with valid details
* Pre-condition: Registration page must be available
* Steps to Execute:
    1. Open E-commerce application
    2. Click Register
    3. Enter valid name, email & password
    4. Click Submit
* Test Data: Valid name, unique email, valid password

* Expected Result: account should be created successfully.

* Status: Pass/Fail.

Test Case ID: ETC02

* Test Case: Login with valid credentials

* Precondn: User must have a registered account

* Steps to Execute:

1. Open login page
2. Enter registered email & correct password
3. Click Login

* Test Data: Valid email & password

* Expected Result: User should log successfully, home page displayed

* Status: Pass/Fail.

Test Case ID: ETC03

* Test Case: Search for existing product

* Precondition: Products must available in Beaton

* Steps to Execute:

1. Open Application
2. Enter product name in search bar
3. Click Search

* Test Data: Laptop

* Expected Result: Relevant products should displayed

* Status: Pass/Fail.

Test Case ID: ETC04

* Test Case : View Product details
* Precondition : Product must exist in catalogue
* Steps to Execute :
    1. Search for product
    2. Select required product
    3. Open details page
* Test Data : An existing product
* Expected Result : Correct product name, descrip, price be displayed
* Status : Pass / Fail

Test Case ID: ETC05

* Test Case : Add product to existing shopping cart
* Precondition : Product must be available in Stock
* Steps to Execute :
    1. Open product page
    2. Click Add to cart
    3. Open Shopping Cart
* Test Data : Available product, Quantity 1
* Expected Result : Selected product should appear in cart.
* Status : Pass / Fail

5) E-Commerce Application - Negative Test Scenario & Test Cases

Negative Test Scenario:

Test Case ID: NTC01

* Test case: Register with invalid email address
* Precondition: Registration page must be available
* Steps to Execute:
    1. Open Registration Page
    2. Enter name & invalid email address
    3. Enter valid Password
    4. Click Submit
* Test Data: Email abc123.
* Expected Result: Application should reject email & displays message
* Status: Pass/Fail

Test Case ID: NTC02

* Test Case: Login with incorrect password
* Precondition: User must have registered account
* Steps to Execute:
    1. Open login page
    2. Enter Registered email
    3. Enter incorrect password
    4. Click Login

* Test Data : Registered email & incorrect password.

* Expected Result: Login should fail, appro error message displayed

* Status : Pass/Fail

Test Case ID: NTC03

* Test Case : Submit Registration with empty fields

* Recondition : Registration page must be displayed

* Steps to Execute :

1. Open Registration page

2. Leave Required fields empty

3. Click Submit

* Test Data : Empty Name, email & password.

* Expected Result: Application shld display reqr field messages

* Status : Pass/Fail